

Stage 3 Judges' Rubric - DEMI Grade 5

Arduino + AI Project - Stage 3 Judge Evaluation

Team Name / #	Project Name	Judge Name	Date

How to Score Every Row Enter ONE number from 1 to 5. Every row is worth 5 points; harder sections have more rows and therefore more weight.

1	Weak	Missing, incorrect, non-functional, or little/no meaningful evidence.
2	Developing	Partially demonstrated; important gaps remain or substantial support is needed.
3	Meets Expectations	Core requirement is present, understandable, and works at an acceptable/basic level.
4	Strong	Complete, accurate, well integrated, and supported by good evidence; only minor issues.
5	Excellent	Highly effective, reliable, purposeful, well-tested/polished, and clearly justified by the team.

Judge the final Stage 3 work, using the previous-phase requirements as the baseline. **Flow:** reward evidence, correct integration, and purposeful design - not complexity for its own sake. **Testing:** look for real trials/results, not only claims. **Hardware:** reward the best-fit use of components and reliable integration. **Teamwork & presentation:** look for shared ownership, clear communication, and the ability of multiple members to answer questions.

Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Problem, Research & Solution			Section Weight: 15 points
Problem, target user & research	Clearly explains the problem, user, importance, and supporting research about similar ideas or needs.		
Solution alignment	The proposed solution and its functions directly address the researched problem.		
User value	The team can explain how the final system helps the target user and why the solution is worthwhile.		
Hardware & System Design			Section Weight: 25 points
Purposeful kit/component choice	Hardware is selected because it fits the solution; unnecessary components are avoided and each chosen part has a clear role.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Hardware & System Design (continued)			Section Weight: 25 points
Input/output component use	Sensors and outputs/actuators are used correctly and their roles match the Stage 2 hardware plan.		
Circuit correctness & safety	Connections are technically appropriate, organized, and safe for the intended prototype.		
System integration	Input -> Arduino/controller -> output behavior works as one coherent system and matches the block/circuit design.		
Best use of hardware & reliability	Hardware capabilities are used effectively for the problem, and the physical setup performs consistently during repeated demonstrations.		
Programming & AI Integration			Section Weight: 20 points
Programming logic	Events/inputs, variables, conditions, loops, and outputs work together correctly.		
Code organization	Functions or reusable blocks are used appropriately; code is readable and avoids unnecessary repetition.		
AI/ML feature relevance	AI feature supports the same problem and is more than a disconnected add-on.		
AI evidence & integration	Team shows model/concept evidence and clearly explains how AI input/output connects to the overall solution.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Prototype, Testing & Improvement			Section Weight: 15 points
Prototype completeness	The physical/digital prototype demonstrates the main planned behavior end-to-end.		
Evidence of testing	Shows repeated tests, observations, measurements, screenshots, logs, or other evidence rather than only stating that testing happened.		
Improvement & reliability	Team uses test findings to improve timing, thresholds, interaction, accuracy, or reliability and can explain the change.		
Business & Value			Section Weight: 10 points
Customer & value proposition	Target customer/user matches the original problem and the value is clear.		
Practical delivery / feasibility	Explains a realistic way the solution could be offered, used, or adopted.		
Teamwork & Presentation			Section Weight: 15 points
Shared teamwork & contribution	Roles are balanced and all members visibly contribute to building, testing, or explaining the project.		
Technical communication	Team explains hardware, code, AI, and design decisions in language appropriate to judges.		
Presentation, demo & Q&A;	Presentation is clear and well paced; demo evidence is convincing; answers show shared understanding.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Extra Requirements / Optional Bonus			Optional Bonus - max 10
Advanced programming feature	Optional stronger programming feature (functions, lists, nested logic, etc.) is useful and supported by code/logic evidence.		
Advanced AI feature	Optional AI feature goes beyond the basic AI requirement and is supported by a model, screenshot, result, or credible prototype evidence.		

Score Summary

Section	Maximum	Section Subtotal	Notes
Problem, Research & Solution	15		
Hardware & System Design	25		
Programming & AI Integration	20		
Prototype, Testing & Improvement	15		
Business & Value	10		
Teamwork & Presentation	15		
CORE TOTAL	100		

OPTIONAL BONUS (record separately, max 10)	
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